Queen Mary
University of London

May/June Examination Period 2026

ECS7005P Risk and Decision-Making for Data Science and AI Duration: 2 hours

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Answer FOUR questions

If you answer more questions than specified, only the first answers (up to the specified number) will be marked. Cross out any answers that you do not wish to be marked

Calculators, including programmable calculators, are permitted in this examination.

Complete all rough workings in the answer book and cross through any work that is not to be assessed.

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Question 1

An object recognition system is applied to a scene in which 45% of the objects are pens, 35% are knifes and 20% are dogs. If the object is a pen, there is a 50% chance that it will be recognised correctly by the system, 80% chance if it is a knife, but only 15% if it is a dog. Suppose an object is selected at random from the scene:

a) Find the probability that the system recognises an object correctly.

[2 marks]

b) If the object was correctly recognised, find the probability that it is:

i) a pen
ii) a knife
iii) a dog

[9 marks]

c) If the object was wrongly recognised, find the probability that it is:

i) a pen
ii) a knife
iii) a dog

[9 marks]

Question 2

A company evaluates the effectiveness of a voluntary job training programme on hiring outcomes. The following causal relationships are known:

- A candidate's **education** level influences whether they enrol in the **training programme**. (Candidates with lower education are more likely to enrol.)
- **Education** also directly affects **hiring** success**.** (Higher education increases hiring probability.)
- The **training programme** directly improves **hiring** success. (Training increases hiring chances.)

a) Using the description above, draw a Bayesian network. Clearly label all nodes and directed edges

[10 marks]

b) Recruiters make hiring decisions based only on candidates' skills and education, not on whether they took the training. However, when the company compares overall hiring rates, candidates who took the training appear less likely to be hired than those who did not.

i) Explain what Simpson's paradox is.

ii) Explain why Simpson's paradox can occur in this training study.

iii)Name the bias illustrated in this scenario.

iv)What happens if we don't control for this bias?

[14 marks]

c) A hiring AI is trained on historical data containing only "Training participation" and "Hiring outcome". Education is not included.

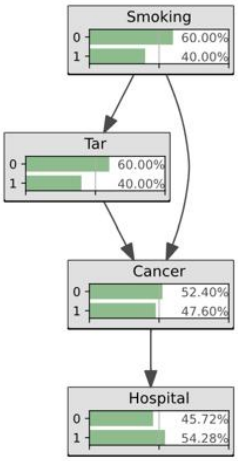
i)Explain one harmful consequence this AI could produce.

ii) Suggest a way to prevent this issue.

[6 marks]

Question 3

A hospital is studying lung cancer using the Bayesian network illustrated in Figure 1, which includes the associated conditional probability tables (CPTs). Given that a patient is a smoker and has lung cancer, we examine the updated Bayesian network after incorporating this evidence, shown in Figure 2. Use these networks to answer the following questions.



	Tar	
Smoking	0	1
0	0.8000	0.2000
1	0.3000	0.7000

		Cancer	
Smoking	Tar	0	1
0	0	0.8000	0.2000
	1	0.1000	0.9000
1	0	0.6000	0.4000
	1	0.2000	0.8000

Figure 1

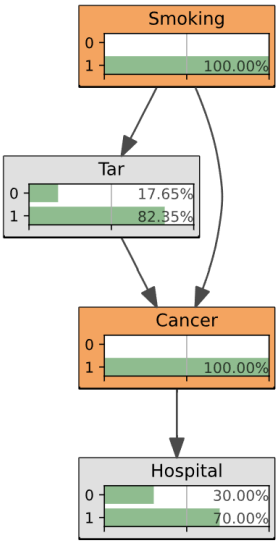


Figure 2

a) Consider the variable Tar in lungs (T=1). Identify which is (i) the prior (before entering any evidence),(ii) the likelihood of observing cancer when T=1, and the patient is a smoker, and (iii) the posterior probability in Figure 2. (specify notation and probability).

[6 marks]

"Continues on next page..."

b) Imagine that we are looking at a new patient who is a smoker, and we implement a treatment that removes tar from the lungs. Explain clearly (but briefly) how you would use the provided Bayesian network (Figure 1) to find the likelihood that the patient has cancer. Describe the steps required and draw the resulting Bayesian network, indicating any variables that must be observed.

[8 marks]

c) Suppose you are looking at the patient represented in Figure 2. You are wondering whether this patient would still have developed lung cancer if a tar-removal treatment had been applied. Explain the reasoning, draw the Bayesian network structure (including any observations), and explain clearly all the steps that you need to follow to answer that hypothetical question.

[10 marks]

d) Could questions (b) and (c) be correctly answered using a standard regression model instead of the Bayesian network? Briefly justify your answer.

[6 marks]

Question 4

Are the interpretations below correct or wrong, and why?

a) Consider the relationship between outdoor temperature and ice cream sales. A negative correlation between these variables would indicate that colder temperatures lead to lower demand for ice cream.

b) Suppose there's a positive correlation between the number of umbrellas sold and the incidence of flu cases in a city. This correlation suggests that buying more umbrellas can cause more people to get the flu.

c) You set a significance level (alpha) of 0.05 before conducting your hypothesis test. After analysing the data, you obtain a p-value of 0.06. An interpretation would be that the observed result is statistically significant at the 0.05 level, and you reject the null hypothesis accordingly.

d) Suppose you conduct a study on the relationship between exercise and weight loss and obtain a p-value of 0.25 (significance level is 0.05). This non-significant result indicated that there is no relationship between exercise and weight loss.

e) Suppose a study calculates a 95% confidence interval for the average weight loss due to a new diet to be [5 kg, 7 kg]. This means that if we were to repeat the study many times, approximately 95% of the resulting confidence intervals would contain the true average weight loss.

[20 marks]

End of Paper